RESEARCH REPORT · VITKI

Methods report, partially tested

The scan as a clue: with the 3D traces the language model filled in 60 percent of the hidden words correctly instead of 54, in a test where the answer was not hidden from the whole chain

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Timeline of the study: The trace reader was built and tuned on Sö 113 17 October 2026 · A first run on Sö 113 and Sö 134 was made 17 October 2026 · The main test was run 17 October 2026 · The test on real lacunae was run 17 October 2026 · The test was recomputed without Sö 113 31 October 2026 · The saved rows were recounted for the correction 5 November 2026

ABSTRACT

What has not been tested. The answer was not hidden from the whole chain: the scan's runes were paired with text positions using the full known text, the hidden word included, and the pairing beat chance only on Sö 134, where the share of right words did not change. The test concerns the groove-5 measurement method and the textmodel-17 language model; Vitki now (5 November 2026) uses groove-7 and textmodel-19, which have not been tested in the same way. Nor was the test locked in a plan, and the trace reader's settings were derived on one of the stones in the test. The test hides whole words of whole runes; at real damage the traces have been tested on only five stones, and there fewer suggestions agreed with the editors' reading with the traces than without them.

Background and aim. When runes are damaged the runologist reads what remains on the stone, and a language model can suggest what stood there. The question here is whether what a 3D scan shows of the damaged runes makes the language model's suggestions better.

Material. 17 runestones in Södermanland that Laila Kitzler Åhfeldt has scanned in 3D and published openly, with their text in the Scandinavian Runic-text Database (Rundata, 2014 version with RUNDATA.xls updated 2018); 157 words that could be tested.

Method. A trace reader reads in the scan's depth image which of the sixteen runes each detected rune can be, with a probability for each. The scan's runes are paired with the positions of the text. Every

whole word of two to six runes is hidden, and a language model trained without the stone fills it in: without traces, with the traces and with the same traces shuffled at random among the stone's positions. The comparison is paired, word by word.
Results. With the traces 59.9 percent of the words came out right, against 53.5 percent without them (10 words became right with the traces and none became wrong; the probability of the right word rose, $p = 3.7 \cdot 10^{-6}$). With shuffled traces the gain vanished and the probability fell ($p = 0.0088$). Without the stone on which the settings were derived: 63.4 against 57.2 percent. On Sö 134, the only stone where the pairing beat chance, 71.4 percent of the 14 words came out right in all three conditions. On five stones with real lacunae 3 of 7 suggestions agreed with the editors' reading in Rundata without traces and 2 of 7 with them.
Conclusion. The traces raised the probability of the right word in the test, but since the pairing saw the answer, what the scan reads cannot be told apart from what the pairing already knew, and on the only reliably paired stone the share of right words did not change. The method needs to be tested again with the answer hidden from the whole chain and with the versions Vitki uses. At severe damage the traces should be shown to the reader and weighed in only once the reader has confirmed them.

Keywords: *runestones, Viking Age, 3D scanning, damaged runes, language model, lacunae, restoration, Södermanland*

HIGHLIGHTS

- What a 3D scan shows of a rune raised the probability of the right word in the test, but the pairing between scan and text saw the answer, so the test does not show how much the scan helps when the answer is unknown.
- The idea that the scan and the language read a runestone together, one showing what can be seen and the other what can be said, remains to be tested with the answer hidden from the whole chain and with the versions Vitki uses.
- At severe damage the traces can mislead, so a human must see and confirm them first.

1. Introduction

Many runic inscriptions have damaged or lost runes. The runologist reads what remains on the stone, and the editions mark certain, uncertain and lost runes. In recent years language models have been used to suggest what stood in damaged inscriptions; the best known is Ithaca for Greek inscriptions (Assael et al. 2022). Laila Kitzler Åhfeldt has scanned a large number of runestones in 3D to study carving technique (Kitzler Åhfeldt 2002; 2015) and published the scans openly (Kitzler Åhfeldt 2024).

What is missing is a way of letting what the scan shows of a damaged rune steer what a language model suggests, and a test of whether it helps. The aim here is to report a trace reader that does so and to test it on hidden words on scanned stones.

2. Pre-registration: what was fixed in advance and what was not

The test was not locked in a plan. The trace reader was built in steps on 17 October 2026. Its two settings, how sharply the probabilities are distributed and what an extra part of a rune costs, were tried out on Sö 113 alone and then validated on Sö 134 and Sö 207, which had not been used. The main test was then fixed in the code, with three conditions and the same measures for all stones. A first run the same day on Sö 113 and Sö 134 (26 words) gave the same results without traces and with the traces, but with shuffled traces the suggestion differed for 3 of the 26 words, all on Sö 134, because the shuffle was drawn differently when the two stones were run together; 2 of them came out right in the final run but wrong in the first. The final run was made stone by stone on all 17 stones, and it is the one reported. Sö 113 is among the 17 stones; on 31 October the test was recomputed without Sö 113, with the same test on the saved rows. The test on real lacunae was run the same day as the main test.

3. Material

- **The scans.** Laila Kitzler Åhfeldt, Swedish National Heritage Board, scanned the runestones of Södermanland with an ATOS scanner in the project Runristandets dynamik and published them on Zenodo under CC BY 4.0 (Kitzler Åhfeldt 2024). Of these, the 17 stones were used for which Rundata has a word text that can be paired with the detected runes. Runestones only.
- **The texts.** Transliteration and normalisation from the Scandinavian Runic-text Database (Rundata), 2014 version with RUNDATA.xls updated 2018.
- **The words.** Every whole word of two to six runes: 197 word positions, 157 of them where all three conditions gave a result.
- **Real lacunae.** Five stones from the same series with lacunae marked in Rundata: Sö 162, Sö 206, Sö 211, Sö 360, Sö 45.

4. Method

4.1 The trace reader. In the scan's depth image the runes' staves are found. Parts of the groove outside the staves, branches and bows, get a weight by how deep, elongated and long they are, since weathering and marks in the surface give shallow and short traces. Each detected rune is compared with the shape of each of the sixteen runes, with a cost for parts missing or too many, and the costs are turned into a probability per rune. The grooves were measured with the version of the measurement method in use on 17 October 2026 (groove-5); Vitki has since moved to a later version (groove-7), which has not been tested in this way.

4.2 The language model. The project's model of runic text, in the version in use on 17 October 2026 (textmodel-17), suggests words and weighs them by how common they are in the runic texts and in their context. It is trained without the stone being tested. The traces are put in softly: a rune the scan does not support becomes less probable, not impossible. In the main test each such rune

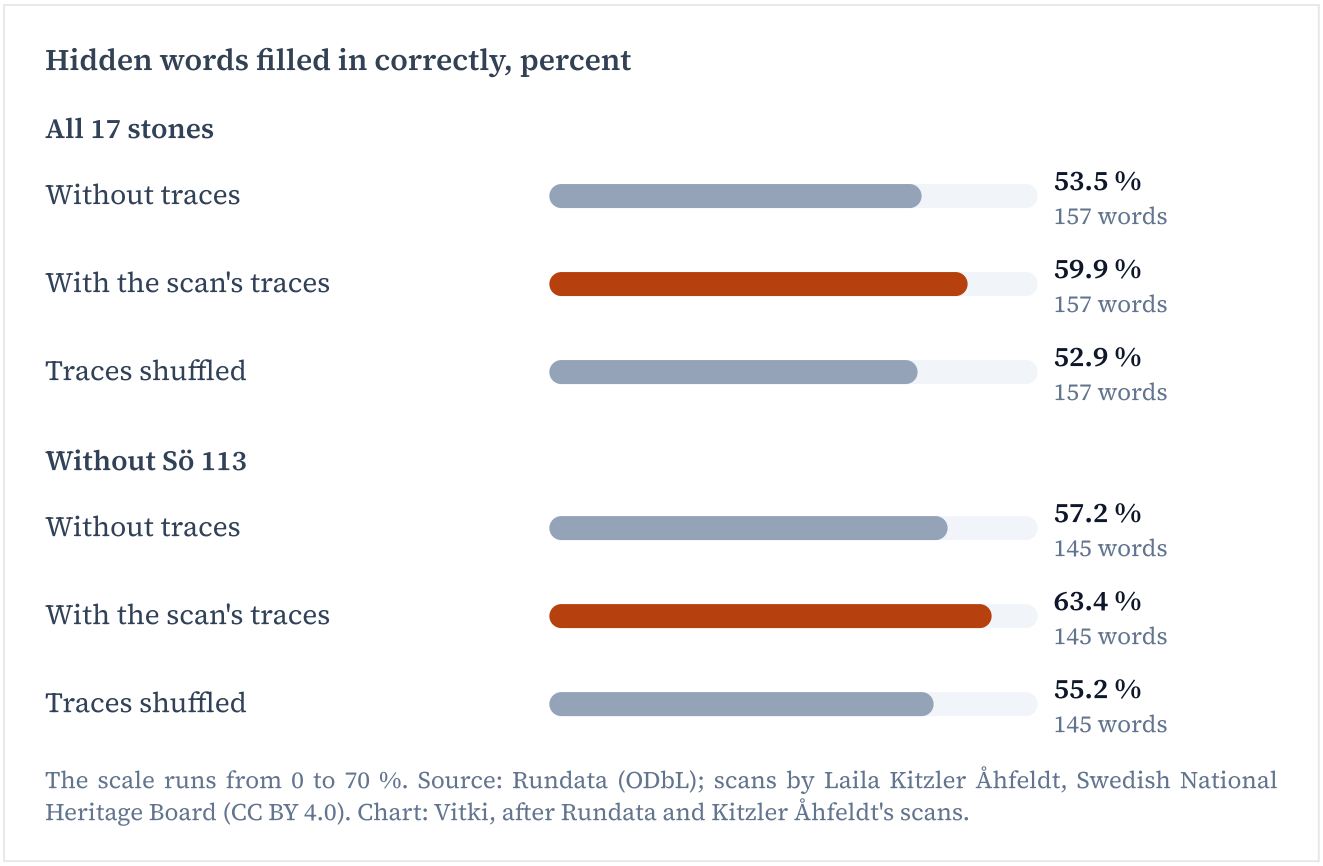
weighed a suggestion down by a factor of 0.2, in the test on real lacunae by 0.05, the factor Vitki also uses.

4.3 The test. Every whole word of two to six runes is hidden and filled in under three conditions: without traces, with the trace reader's probabilities for the word's runes, and with the same sets shuffled at random among the stone's positions. The last condition was meant to test whether the traces help because they read the stone, or merely because they reduce the number of possible runes. The measures are the share of words that came out right and the probability of the right word; the probability is compared word by word with Wilcoxon's test for paired values, the share of right words is reported without a test. To know which detected rune belongs to which position in the text, the scan's runes were paired with the text's runes, once per stone and using the full known text, before any word was hidden (see 4.4). The pairing was tested against the same pairing with the text shuffled 20 times, and counted as reliable when the share of paired runes whose text rune was among those the scan supports lay at least two standard deviations above that of the shuffled text.

4.4 Deviations and errors. No plan existed, so no deviations from a plan can be reported. Three things were not reported in the first two versions of the report: that the pairing saw the hidden word, that the code's own check found the pairing reliable on only one stone, and that the test was run more than once. The pairing of the scan's runes with the text used the full known text, including the word that was then hidden, and it costs nothing where the text's rune is among those the scan supports; it can therefore pull the scan's sets to the positions where they agree with the answer. This can favour the condition with traces, and the control with shuffled traces does not separate that advantage from the traces' own reading, since both vanish when the sets are moved. The code's own check found the pairing reliable only on Sö 134 (1 of 17 stones). The error has not been corrected and the test has not been rerun. The first run is reported in 2. The recomputation without Sö 113 was made after the test.

5. Results

5.1 The main test. 157 words on 17 stones. Right word: 53.5 percent without traces (84 words), 59.9 percent with the traces (94) and 52.9 percent with the traces shuffled (83). With the traces 10 words came out right that were wrong without them, and no word that was right became wrong; with shuffled traces 4 words became right and 5 wrong. The probability of the right word: 0.379 without, 0.414 with and 0.365 with shuffled traces. The traces gave a higher probability than no traces, $p = 3.7 \cdot 10^{-6}$, and higher than shuffled traces, $p = 2.1 \cdot 10^{-8}$. Shuffled traces gave a lower probability than no traces, $p = 0.0088$; in the share of right words the difference was one word of 157. All p-values concern the probability of the right word, not the share of right words. In plain words: with the scan's traces about six hidden words in ten came out right instead of just over five; 10 words became right and none became wrong. When the traces were shuffled the gain vanished. But the pairing between scan and text saw the answer (4.4), so some or all of the gain may come from the pairing rather than from the scan.



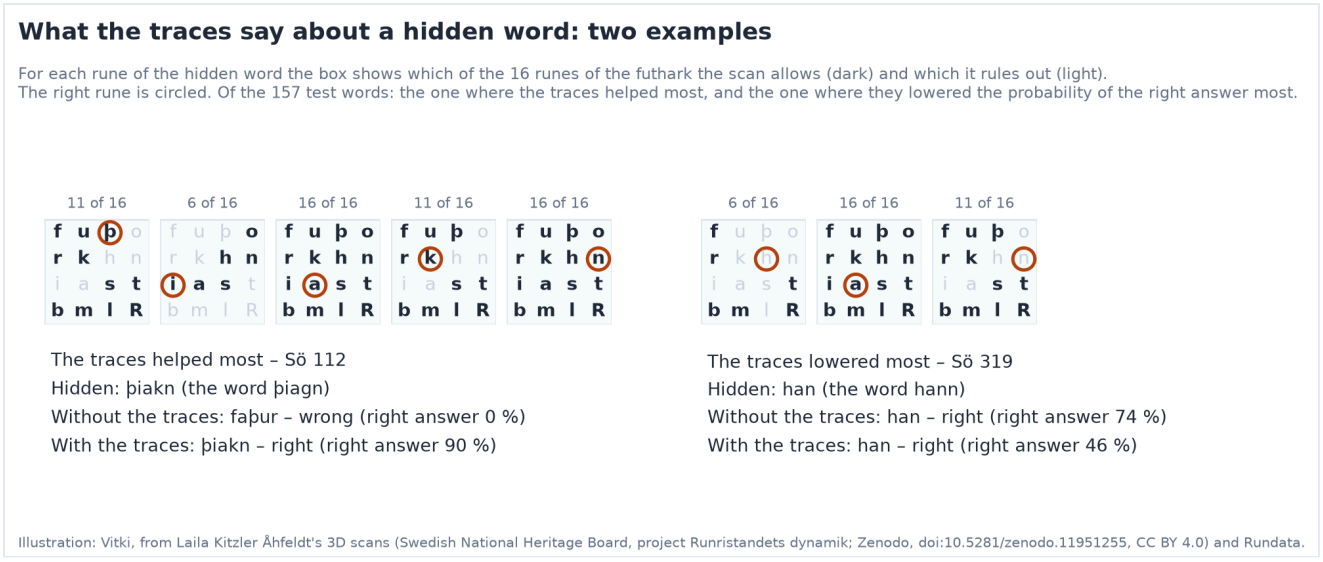


Figure 1. What the traces say about a hidden word. For each rune the boxes show which of the 16 runes of the futhark are in the scan's set (dark) and which are outside it (light); the right rune is circled. The image calls the light runes ruled out, but in the model they only become less probable: in the test each such rune weighed a suggestion down by a factor of 0.2. Left, the word where the traces helped most (Sö 112); right, the word where they lowered the probability of the right answer most (Sö 319), where the right rune is outside the set in two of three places. In none of the 157 words did the traces make a right word wrong; in 10 they made a wrong word right. Illustration: Vitki, from Laila Kitzler Åhfeldt's 3D scans (Swedish National Heritage Board, project Runristandets dynamik; Zenodo, CC BY 4.0) and Rundata.

5.2 Without Sö 113, and the only reliably paired stone. Without the stone on which the settings were derived: 145 words on 16 stones, 57.2 percent right without traces, 63.4 with and 55.2 with shuffled; the probability of the right word with the traces against no traces $p = 5.9 \cdot 10^{-6}$, against shuffled $p = 1.4 \cdot 10^{-8}$. Only on Sö 134 did the pairing beat chance. There 71.4 percent of the 14 words came out right in all three conditions, and no word changed; the probability of the right word rose with the traces (on average from 0.45 to 0.50), $p = 0.051$ against no traces, just above the usual 0.05 limit and with 14 words a weak sign, and $p = 0.0012$ against shuffled.

5.3 Calibration. Not tested. The test saved the probability of the right word but not that of the model's first choice, so it cannot show whether the model's probabilities match how often the suggestions came out right. The 7 words in real lacunae are too few for such a test.

5.4 The earlier step, on undamaged runes. Before the language model was connected, the trace reader was tested on its own on the detected runes of Sö 113, Sö 134 and Sö 207: the probability of the right rune was on average 0.077 against 0.066 when the text was shuffled (200 shuffles, $p = 0.010$), and the right rune came first in 19.3 percent against 13.8 ($p = 0.015$). The traces alone say little.

5.5 Real lacunae. On the five stones with lacunae marked in Rundata (7 words with lost runes) the chain from scan to suggestion was run, and the suggestions were compared with the editors' reading in Rundata. Without traces 3 of 7 words agreed, with the traces 2 of 7. Where the staves are whole the traces kept the right words, *stæin* on Sö 211 and *gæiRfastr* on Sö 45. On Sö 162 the traces pulled the suggestion from *stæin* (stone), which agrees, to *staf* (staff), which does not; the code's rule for the same word counted *staf* as *stæin*. On Sö 206 Rundata reads the name *G[uðl]aug*, and both the suggestion without traces, *karl*, and the one with traces, *kumbl*, are wrong. On Sö 360 the traces lowered the probability of the suggestion *farinn* from 0.82 to 0.27; the editors leave that rune sequence uninterpreted. The trace reader is tested on undamaged runes, and weathering at a lesion produces exactly the false branches it then reads.

6. Discussion

What the method can do. In the test the scan's traces raised the probability of the right word, and the same sets in the wrong place removed the gain. But the pairing between scan and text saw the answer, and on the only stone where the pairing was reliable the share of right words did not change. The test thus shows that the chain from scan to suggestion can be built, not how much the traces help when the answer is unknown. It is a first step towards 3D scanning and a language model reading an inscription together, the scan showing what can be seen and the model what can be said.

Against earlier research. Ithaca uses only the text (Assael et al. 2022). Here what can be seen on the stone is added, as probabilities the model weighs in. The scans were made to study carving technique (Kitzler Åhfeldt 2015) and may also become a source for reading.

Weaknesses, in order of weight.

1. **The answer was not hidden from the pairing.** The scan's runes were paired with the text with the hidden word visible, and the control with shuffled traces does not separate the pairing's advantage from the traces' own reading. The pairing beat chance only on Sö 134, and there the share of right words did not change.
2. **Older versions.** The test concerns the groove-5 measurement method and the textmodel-17 language model, in the main test with a factor of 0.2 for the traces. Vitki now (5 November 2026) uses groove-7 and textmodel-19 with a factor of 0.05; they have not been tested in the same way.
3. **Hidden words are not damaged words.** The test hides words whose runes are whole on the stone. At real damage the traces have been tested on only five stones, and there 2 of 7 suggestions agreed with the traces against 3 of 7 without.
4. **No locked plan.** The test's conditions were fixed before the run, but without a plan with a checksum, and a first run gave partly different suggestions with shuffled traces. The settings were derived on Sö 113, which is in the test; without it the result is the same.
5. **One scanning series.** All stones come from the same series in Södermanland, with the same scanner. Other scanners and other stones may give other traces.
6. **The measure of a right word.** A word counts as right when it is the same word or inflection; the rule sometimes counts neighbouring forms as the same, which applies equally to all three conditions. In the test on real lacunae a similarity rule counted *staf* as *stæin*.

Where the method should not be used. On its own, at severe damage. There the traces should be shown to the reader of the stone and weighed in only once the reader has confirmed them; that is how Vitki was built when this report version was written (5 November 2026).

What would overturn the method. If a new test with the answer hidden from the pairing as well gave no improvement, if the traces did not help on stones from other scanning series, or if they misled more often than they helped at real damage in a larger material.

7. Conclusion

With the 3D scan's traces the language model filled in 59.9 percent of the hidden words correctly against 53.5 percent without, but the pairing between scan and text saw the answer, so the test does not show how much the traces help when the answer is unknown. The test concerns older versions of the measurement method and the language model and needs to be redone with the answer hidden from the whole chain. At severe damage a human must confirm what the scan sees before it is weighed in.

Data and code

The test was run on 17 October 2026 (commits a71c134 and f3b8f01) with the groove-5 measurement method and the textmodel-17 language model; the 31 October recomputation only re-aggregated the saved rows. When this report version was written (5 November 2026), Vitki used the groove-7 measurement method (the trace reader has read the groove-7 measurements since com-

mit add12ef, 3 November 2026) and the textmodel-19 language model with a factor of 0.05 for the traces; neither has been tested in this way. The code state is given by commit in Precatio/runes, since no release of Vitki with a DOI existed when this report version was written. The check for report version 3 (`kontroll_v3.py`) and the builder of that version are in commit 6eae0af (5 November 2026). The code and the result files below are openly available on GitHub under GPL-3.0.

- The trace reader and the tests: `utdata/sparlasare/` (the steps `led_a.py` – `led_g.py` ; the main test `led_f.py` with the rows per stone `led_f_Sö_*.json` , the scan's measured features `*_feats_e.json` and the summary `led_f_summering.py` → `led_f_summering.json` ; the first run `led_f.json` ; the earlier step `led_e.py` → `led_e_perm.json` ; the recomputation without Sö 113 `utan_so113.py` → `utan_so113.json` ; real lacunae `led_g.py` → `led_g.json` ; the check for report version 3 `kontroll_v3.py` → `kontroll_v3.json`).
- The grooves: `src/auto_grooves.py` (groove-5 at commit a71c134). The language model: `src/runlm.py` and `src/textmodel.py` (textmodel-17 at commits a71c134 and f3b8f01). The trace reader in Vitki: `src/trace_reader.py` .
- The texts: Rundata (version 2014, RUNDATA.xls 2018), `data/rundata.json` , unchanged since 4 October 2026 (git blob ee37aec), the same file the test read.
- The scans: Kitzler Åhfeldt 2024, doi:10.5281/zenodo.11951255.
- The report is built by `utdata/blogg/build/build_rapport_sparlasare.py` , which reads every number from the result files and stops if anything disagrees.
- In Vitki the traces are shown in the stone analysis under "Vad skannern ser i luckan" (what the scanner sees in the gap), and they count only once the user has confirmed them.
- The statistical terms are explained in the glossary: vitkirunor.se/en/forskningsrapporter/begrepp.

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Scandinavian Runic-text Database (Rundata), 2014 version with RUNDATA.xls updated 2018. Department of Scandinavian Languages, Uppsala University. Open Database License.

Change log

- **Version 1, 2026-10-31.** First published version.

- **Version 2, 2026-11-01.** Figure 1 added: what the traces say about a hidden word, in two examples chosen by a rule given in the caption. No numbers, interpretations or conclusions have changed.
- **Version 3, 2026-11-05.** Corrections after a review before archiving in Zenodo; no result file of the test has changed and nothing has been rerun. The abstract now begins by saying that the answer was not hidden from the whole chain (the pairing of the scan's runes with the text saw the hidden word, and it beat chance only on Sö 134, where the share of right words did not change) and that the test concerns groove-5 and textmodel-17, not the versions Vitki uses; the title, the method, the deviations, the results, the discussion, the conclusion and the essentials box say the same. Data and code give the test's commits and versions: the sentence about changes up to 31 October was wrong, since the recomputation only re-aggregated saved rows, and the sentence about a future release has been replaced by the code state. The p-values concerned the probability of the right word but stood next to the share of right words; with shuffled traces one word fewer came out right than without traces (83 against 84 of 157), so worse than no traces holds only for the probability. The test was not run once: the first run is now reported. Real lacunae: on Sö 206 both karl and kumbl are wrong, and staf on Sö 162 does not agree with stæin; 3 of 7 suggestions agreed with Rundata without traces and 2 of 7 with them. Section 5.3 Calibration (not tested) has been added and the following sections renumbered; the sharp test is now called the main test; the figure caption says that runes outside the scan's set become less probable, not ruled out; a reference that was not cited has been removed; the chart credits Vitki; the glossary is linked. The new numbers were counted in the saved rows with a new check script.
- **Version 4, 2026-11-05.** A second review before archiving in Zenodo checked what version 3 changed; no result file has changed and nothing has been rerun. 5.1 now says that some or all of the gain may come from the pairing; version 3 said part of it, although the test cannot show that any part comes from the scan. In 5.2, p = 0.051 on Sö 134 is explained in plain words, with the mean probabilities 0.45 and 0.50. The abstract gives Rundata's version in the same way as the rest of the report (2014 version with RUNDATA.xls updated 2018); version 3 gave a different version designation there. The change log has the line for version 1, and Data and code gives the commit of the check and the builder of version 3 (6eae0af). Changes that version 3 made but did not log: Rundata's version was given in the material, in Data and code and in the references; 5.4 no longer says that the traces say more together with the language; the discussion no longer says that the scans turn out to be a source for reading, only that they may become one; and the discussion and Data and code say that the traces count in Vitki only once the user has confirmed them, as Vitki was built when version 3 was written.

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Cite as

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